

Project Design Phase

Solution Architecture

Date	11-10-2025
Team ID	SWUID20250177148
Project Name	SB Works
Maximum Marks	4 Marks

Overview

The solution architecture for SB Works is designed as a robust, efficient, and scalable platform that connects clients with skilled freelancers. Based on a client-server model, the architecture prioritizes transparency, security, and a streamlined workflow. It uses the MERN stack (MongoDB, Express.js, React, Node.js) to ensure real-time communication, efficient data exchange, and a seamless user experience for all parties involved.

The primary architectural goals include:

- Efficient project posting for clients and a seamless bidding process for freelancers.
- A unified workflow for collaboration, project submission, and feedback, all within the platform.
- Secure user management with distinct roles (Client, Freelancer, Admin) overseen by a dedicated team.
- Reliable, real-time communication and notifications to keep users informed.

Solution Architecture Diagram

The technical architecture is composed of three main tiers: a frontend client, a backend server, and a database.

1. Frontend (Client-Side)

- This tier handles the user interface and presentation layer.
- Technologies:
 - UI (React): A JavaScript library for building the interactive user interface and components.
 - Socket.IO: Enables real-time, bidirectional communication, which is essential for features like live chat.
 - Axios: Facilitates communication with the backend through RESTful APIs.
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2. Backend (Server-Side)

- This tier manages all server-side logic, handles requests, and communicates with the database.
- Technologies:
 - Node.js: A JavaScript runtime environment that executes server-side code.
 - Express.js: A web application framework for Node.js that simplifies the creation of robust APIs and manages routing.

3. Database

- This tier is responsible for the persistent storage and retrieval of all application data.
- Technologies:
 - MongoDB: A scalable NoSQL database used to store data such as user profiles, projects, and applications.
 - Mongoose: An Object Data Modeling (ODM) library that provides a straightforward, schema-based solution to model application data and interact with MongoDB.

Solution Architecture Diagram SB Works



